

5.2 Mobile Services And Standards

5.2.1 1G

This is the first-generation telecommunication standard for mobile phones. It was prevalent in the 1980s and early 1990s and used analogue radio signals. It has been almost phased out.

5.2.2 2G

Successor to 1G, the second-generation standards used digital radio signals. This meant that voice signals could be encoded and multiplexed, resulting in more data being squeezed over the airwaves. The voice quality improved considerably, and the kind of static heard on 1G networks is absent. The digital systems also emit lower power. Lower power means that the devices can be smaller, and that the towers become more inexpensive to set up and maintain.

2G networks comprise mainly TDMA and CDMA networks, depending on the type of multiplexing i.e. the way in which the multiple radio waves are combined into a single stream. TDMA standards can be further divided into GSM, iDEN, and more. Indian mobile networks are primarily 2G.

5.2.3 3G

Third-generation networks deliver faster data speeds over existing 2G networks and also more types of services like broadband access and video telephony. The typical data transfer rates over 3G networks are in the region of 5-10 Megabits per second. Depending on location, 3G networks use either the same spectrum as 2G (in the US) or a different spectrum (in Asia and Europe). 3G-enabled phones are prevalent mostly in Europe and parts of Asia like South Korea and Japan. 3G-derived technologies include HSPA (High Speed Packet Access), WCDMA (Wideband Code Division Multiple Access), and UMTS (Universal Mobile Telecommunication System).

5.2.4 4G

Fourth-generation wireless networks are intended to provide a solution where voice, data, and video can be accessed seamlessly